

COSMIC TEAM

AI-Powered Drag Sail for Autonomous Space Debris Mitigation

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Concept Document

PROBLEM

Thousands of inactive satellites and rocket bodies remain in LEO, accumulating for decades. Without intervention, this leads to:

- Cascading collision risk (Kessler Syndrome)
- Mission safety threats and orbital congestion
- Expensive or unscalable active removal methods

No scalable passive solution currently exists that autonomously prioritizes which debris to deorbit first.

SOLUTION

COSMIC TEAM combines **AI orbital analysis** with a passive Kapton drag sail to autonomously assess risk and accelerate re-entry – **no propellant required**.

AI predicts deorbit lifetime and collision risk
Decision engine selects optimal sail deployment %
Burn-wire releases sail → drag increases → faster decay
100% passive, scalable, propellant-free

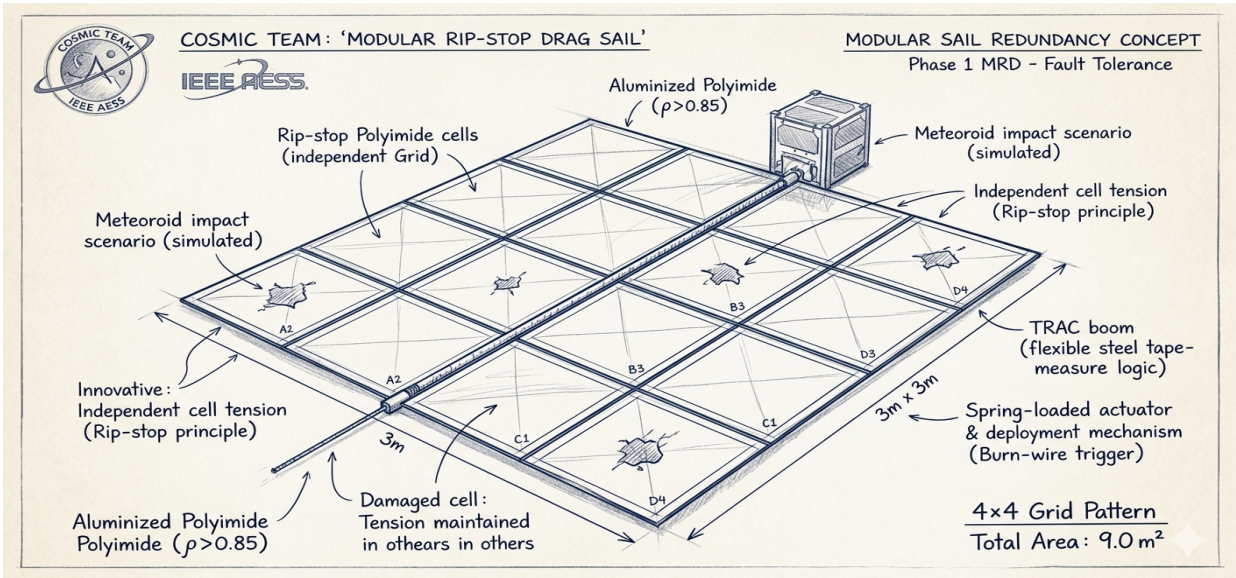
SYSTEM ARCHITECTURE

Orbital Parameters TLE / Space-Track	Feature Engineering 14 orbital features	AI Models A + B RF R²>0.97	Risk & Lifetime Score + years	Decision Engine 5-action output	Sail Deploy
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HARDWARE		AI SYSTEM		
Component	Specification	Model	Purpose	R²
Sail Membrane	12.5 μm Kapton — $p > 0.85$, -269°C to $+400^{\circ}\text{C}$	Model A	Natural Deorbit Lifetime	0.9935
TRAC Booms	Triangular Rollable & Collapsible — compact stow & deploy	Model B	Orbital Risk Score	0.9701
Actuator	Burn-Wire — melts Dyneema → spring releases booms			
Sail Pattern	4x4 Modular Rip-Stop Grid — 9.0 m² total area			

5 Deployment Actions: MONITOR · PREPARE · DEPLOY_PARTIAL · DEPLOY_HIGH · DEPLOY_FULL

Input Features (14): PERIAPSIS, APOAPSIS, BSTAR, ECCENTRICITY, MEAN_MOTION, decay_urgency, bstar_log, RCS_SIZE, OBJECT_TYPE + engineered features



KEY RESULTS — AT A GLANCE

~85% Orbital Lifetime Reduction 25 yrs → 3.4 yrs	0.9935 AI Model R² (Deorbit) MAE = 1.23 years	0.9701 AI Model R² (Risk) MAE = 1.76 pts	13,099 LEO Objects Analyzed Space-Track.org
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PERFORMANCE: BASELINE VS. COSMIC TEAM

Metric	Baseline (No Sail)	COSMIC TEAM	Improvement
Orbital Lifetime	25+ years	~3.4 years	~85% reduction
Collision Risk Period	Decades unmanaged	AI-prioritised deployment	Targeted mitigation
Propellant Needed	Sometimes	None	100% passive
Deployment Trigger	Manual/None	Autonomous AI decision	Scalable

IMPLEMENTATION EVIDENCE Trained Random Forest models (Model A & B) with cross-validation Flask-based interactive dashboard — deployed online MATLAB orbital decay simulation with Earth rotation & burn-up visualization Public GitHub repository with full documentation 13,099 LEO objects processed from Space-Track.org	SUSTAINABILITY IMPACT Reduces LEO object persistence by ~85% per deployed unit Propellant-free — no chemical contamination of orbit AI prioritisation focuses effort on highest-risk objects first Passive & scalable — applicable to CubeSat constellations Directly supports IADC 25-year deorbit guideline compliance
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Limitations & Future Work — Current: simplified atmospheric model, simulation-based deployment assumptions. Next steps: real-time telemetry integration · higher-fidelity astrodynamics · reinforcement learning optimization · on-orbit hardware validation.